

Optimization of Induction of liquid-Liquid Phase Separation of α -synuclein

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1 Abstract

Liquid-liquid phase separation (LLPS) plays a crucial role in organizing biomolecules within cells, facilitating the formation of membranous organelles to regulate cellular processes. Conversely, fibrillation involves the aberrant aggregation of proteins into final oligomeric spherical aggregates and into amyloid fibrils, linked to various neurodegenerative disorders. α -Synuclein, implicated in Parkinson's disease, is among these proteins. LLPS has emerged as a regulator of pathological fibrillation, yet its precise role and link to amyloid oligomers and fibrils formation remains unclear. How the properties of the condensates compare to the aggregated species is also not fully understood.

In this thesis, we sought to induce LLPS in α -synuclein under conditions previously reported in the literature and to investigate its behavior using confocal and conventional microscopy, employing fluorescent labeling to monitor the process. Additionally, the structural characteristics of the protein were analyzed using circular dichroism (CD) spectroscopy and Fourier transform infrared (FTIR) spectroscopy. Additionally the protein purity was tested with Nanodrop spectrophotometry.

However, despite replicating established conditions, the expected phase separation was not consistently observed, with the formation of irregular, often non-fluorescent aggregates occurring instead. Spectroscopic analyses suggested the presence of protein impurities and interference from fluorescent labels and PEG, complicating the interpretation of the results. These findings highlight the challenges of inducing and studying LLPS in α -synuclein and underscore the importance of optimizing protein purity and experimental conditions to accurately investigate its role in neurodegenerative disease progression.

2 Introduction

Liquid-liquid phase separation (LLPS) is a process by which biomolecules can organize themselves into membranellous organelles in the cytoplasm. This compartmentalization is used in regulating a variety of cellular processes and is essential for normal functioning [1-3]. Fibrillation is a process in which undergo misfolding, leading to intermediate oligomers, and then assemble into β -sheet-rich aggregates known as fibrils, leading to the formation of amyloid structures. These protein fibrils and amyloid structures have been associated with various neurodegenerative disorders such as Parkinson's, Alzheimer's and amyotrophic lateral sclerosis (ALS) [4, 5]. α -synuclein has been linked as one of such proteins and as a marker of Parkinson's disease [5]. Meanwhile LLPS has been linked as a regulator of pathological fibrillation as many amyloid peptides have been shown to undergo LLPS before fibrillation [6]. The exact role of LLPS in protein aggregation is not entirely clear and deserves further studies.

2.1 LLPS function and malfunction

In physical terms, liquid-liquid phase separation refers to a transition in density, leading to the creation of a dense phase where molecules can move relatively freely. It is helpful to imagine the phase separation as a droplet with a different make up than its surrounding fluid, a bit like an oil droplet in water. In this two-phase state, a dense phase and a light phase stably coexist. The transition of these densities is commonly depicted using phase diagrams (**figure 1**). A phase diagram serves as a visual representation of the phase behavior of a molecule, achieved by systematically altering conditions. Temperature, concentration, and salt levels are often utilized as parameters in these phase diagrams [7].

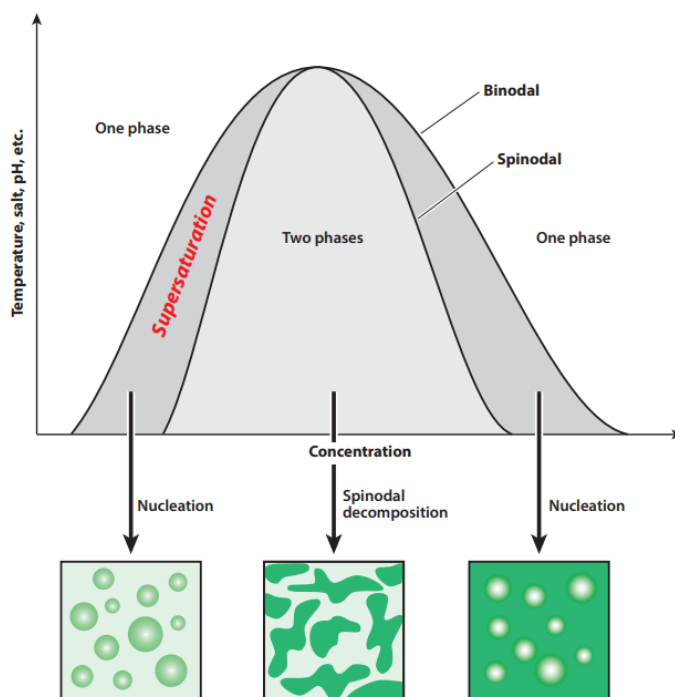


Figure 1. A phase diagram of LLPS. Light green and dark green represent the dilute phase and the dense phase, respectively, in the phase-separated system. Source: [8]

In a phase diagram the binodal, also known as the coexistence line, distinguishes the one-phase state from the two-phase state, marking the region where phase separation is thermodynamically favored. The spinodal delineates the line between nucleation-limited (discrete) phase separation, where nucleation occurs, and diffusion-limited (uniform) phase separation, where spinodal decomposition takes place (**figure 1**) [8].

The increased concentration of proteins and other biomolecules in phase separated condensates allows for various functions such as assembly, sequestration, transportation and certain reactions to occur that would be unlikely or impossible in the cytoplasm [9]. In order to execute these functions the condensates can exchange molecules with the cytoplasm but they resist disassembly by osmosis through certain forces. These forces driving LLPS are those of proteins and RNA's that hold the condensate together with multivalent interactions. These interactions both influence and are influenced by the spatial organization of the molecules within the condensates (**figure 2**) [9, 10].

The concentration of proteins resulting from liquid-liquid phase separation poses inherent risks. Primarily, the elevated protein levels can initiate aggregation processes or cause jamming, resulting in the formation of solid gels or crystals. These solid formations may no longer facilitate the necessary conditions for chemical reactions. To counteract such aggregation, cells employ deaggregases [11] and regulate compartment dynamics through mechanisms like phosphorylation and dephosphorylation [12]. However, in certain circumstances, such as metabolic syndrome or when mutant proteins prone to aggregation are present, cells may struggle to dissolve the aggregates or constrain their growth [7]. Many neurodegenerative brain diseases are characterized by such toxic aggregates such as Alzheimer's, Amyotrophic lateral sclerosis or in α -synuclein's case, Parkinson's disease [4, 5].

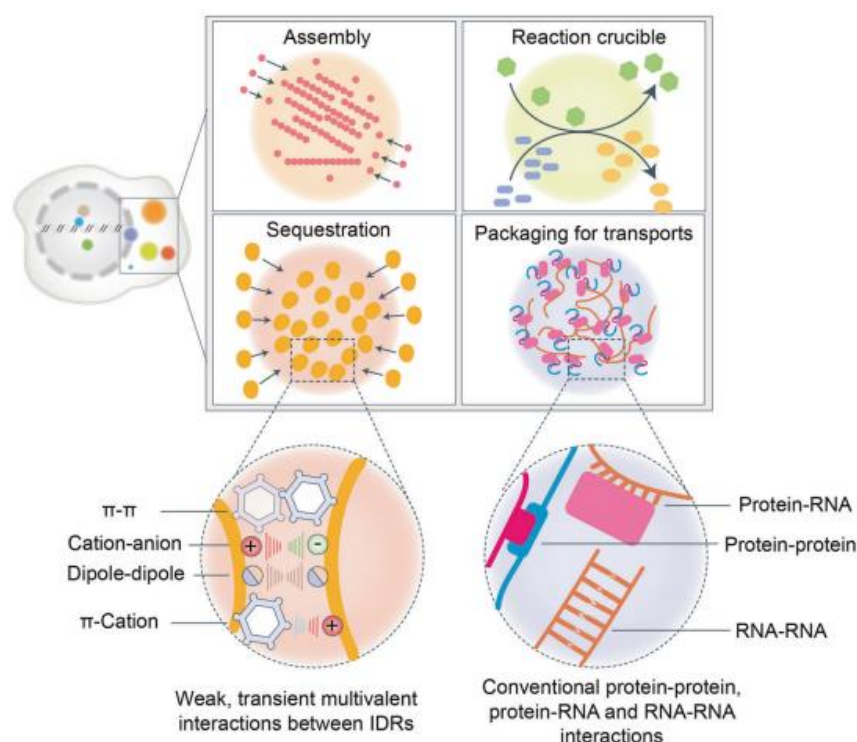


Figure 2. The forces driving LLPS and some of its functions. Functions of LLPS include assembly of large complexes, providing a confined environment that promotes biochemical reactions, sequestering specific proteins to regulate reactions, and acting as packaging for transportation purposes. Additionally, two types of multivalent interactions contribute to LLPS. One type involves conventional multivalent interactions between proteins, proteins and RNA, or RNA molecules themselves. The other type comprises weak and transient multivalent interactions occurring between intrinsically disordered regions (IDRs), which include π - π interactions, cation-anion interactions, dipole-dipole interactions, and π -cation interactions. Source: [9]

Parkinson's disease (PD) presents as a multifaceted neurological disorder with increasing complexity. Traditionally, it has been identified by its hallmark motor symptoms of parkinsonism, linked to Lewy bodies and the loss of dopaminergic neurons in the substantia nigra. However, our understanding now acknowledges PD as a condition with diverse non-motor symptoms of clinical significance. Moreover, its pathology involves not only Lewy bodies but also encompasses various protein aggregates and affects multiple regions of the nervous system and neurotransmitter systems. Besides Lewy bodies, Lewy neurites is another classical neuropathological hallmark of PD. Aggregated α -synuclein is the major component of both these intraneuronal inclusions [13]. While the exact cause of PD remains elusive, the risk of developing it is no longer solely attributed to environmental factors; rather, it appears to result from a complex interplay between genetic and environmental influences impacting fundamental cellular processes. α -synuclein plays a crucial role in many of these cellular processes and its function is impeded by LLPS and aggregation [10, 14-16]. This complexity poses clinical challenges, including difficulties in early diagnosis and symptom management in later stages. Furthermore, current treatments do not effectively halt the neurodegenerative process [17].

2.2 α -synuclein function and malfunction

α -synuclein is a small protein of 140 amino acids that can be subdivided into 3 domains, it is N-terminal and C-terminal domains (NTD & CTD) and a middle non-amyloid- β component (NAC) [10]. The non-amyloid binding domain is the part of the protein that can create the bonds that drive formation of oligomers and fibrils [19]. This part of the protein can be found by looking at β -synuclein. β -synuclein does not form fibrils even though the two proteins are extremely similar. There is a hydrophobic stretch of amino acids in the α -synuclein in residues 71-82 that β -synuclein lacks. When this region is removed α -synuclein loses the ability to form amyloid structures. This shows that this region is responsible for the aggregation [18]. When these aggregates form, they can spread from cell to cell through a cycle involving uptake of external aggregates, co-aggregation with endogenous α -synuclein and exocytosis of the co-aggregates [18].

The role of α -synuclein in normal function is controversial but it is generally understood the protein is essential for the release of neurotransmitters, particularly dopamine, by modulating synaptic vesicle trafficking and exocytosis (**figure 3**) [10, 19]. High expression of α -synuclein (which is associated with disease) inhibits synaptic vesicle exocytosis [20]. These high concentrations in turn can result in α -synuclein misfolding and aggregation in the dopaminergic region. Additionally, in disease conditions the SNARE-dependent membrane fusion is hindered resulting in decreased neurotransmitter release [10]. SNARE stands for soluble *N*-ethylmaleimide sensitive factor attachment protein receptor.

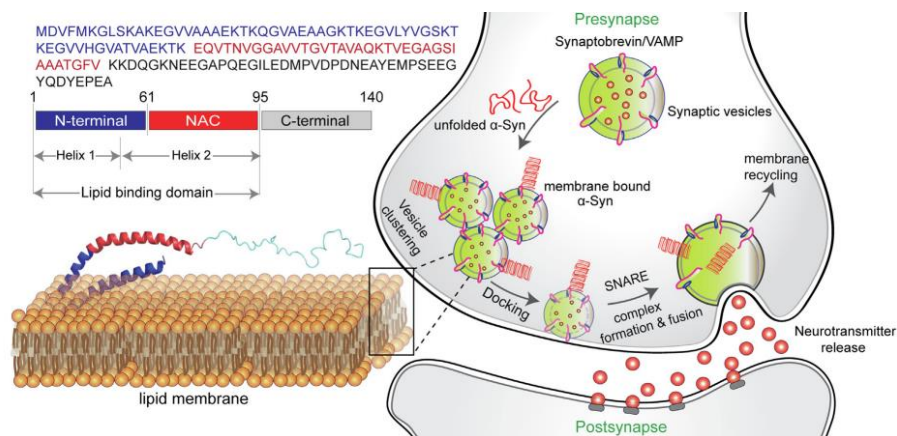


Figure 3. Depiction of α -synuclein's role in neuron vesicle transport. α -synuclein is highly concentrated at the presynaptic nerve terminus and exerts its functional role by binding to the synaptic vesicle (SV) membrane through its N-terminal lipid-binding domain. Upon binding, the initially unfolded protein undergoes a structural transition to adopt an α -helical conformation. This conformational change facilitates SNARE-dependent complex formation. SNARE proteins are important for membrane fusion and cargo transport and together with SNARE regulator proteins they regulate synaptic vesicle fusion. Because of this α -synuclein is crucial for various steps of vesicular trafficking, including SV clustering, docking, and priming [19]. Ultimately, these processes culminate in the release of neurotransmitters at the synaptic junction. Source: [10]

α -synuclein has also been theorized to help in synaptic vesicle (SV) phase separation. It would do this by initiating or helping the phase separation of synapsin, thereby helping the SV cluster organization. This synaptic cluster liquid phase could then in turn act as a stabilizer of α -synuclein that prevents it from aggregating [16].

Besides its interaction with synaptic vesicles and the neuronal membrane α -synuclein has also been found to interact with many other membranes such as the plasma membrane, lipid rafts, the inner nuclear membrane, and mitochondria [21, 22]. Its interaction with mitochondria is interesting because mitochondrial dysfunction is also implicated in PD pathogenesis [23, 24]. α -synuclein/mitochondria interactions are theorized to contribute to neuronal death by increasing oxidative stress and disrupting mitochondrial dynamics. Additionally, α -synuclein can translocate to the nucleus under conditions of increased oxidative stress, potentially exacerbating cellular damage. This develops more quickly in the substantia nigra (the area of the brain where PD occurs) as these cells are more vulnerable to oxidative stress due to their dopamine metabolism producing toxic substances [25].

A-synuclein has also been linked to ferroptosis, a form of iron-dependent cell death characterized by lipid peroxidation. Iron imbalance is closely linked to PD pathogenesis, and ferroptosis involves oxidation of polyunsaturated fatty acyl chains of phospholipids in an iron-dependent manner, leading to mitochondrial swelling and accumulation of lipid peroxides. Studies have demonstrated widespread occurrence of ferroptosis in PD models. In PD, iron accumulation, oxidative stress, and lipid peroxidation are associated with neurodegeneration. Elevated iron levels in the substantia nigra, observed in PD, may result from dysregulation of iron-binding proteins, promoting oxidative stress, α -synuclein phase separation and aggregation, and neuronal death. Under pathological conditions, α -synuclein may directly or indirectly enhance the accumulation of polyunsaturated fatty acids on cell membranes, leading to lipid peroxidation and promoting ferroptosis. Additionally, α -synuclein is an iron-reducing protein and when it aggregates the iron metabolism is disturbed which can also indirectly cause ferroptosis [14].

As α -synuclein aggregates it loses its ability to execute its mentioned functions properly leading to neuron dysfunction and death [26, 27]. The insoluble inclusions that the aggregates form such as Lewy bodies can also cause direct neuronal damage in a gain of function mechanism [22]. Furthermore, cellular compartments and processes are disturbed by α -synuclein. A-synuclein oligomers can for example bind to lipid membranes and disrupt the lipid bilayer [28]. LLPS of α -synuclein has been found to promote its fibrillation [29-35] and conditions that are associated with PD generally induce LLPS as well [36].

There are many more theories, as cells try to get rid of α -synuclein it interacts with many cellular compartments and membranes making it hard to pin down the exact reason for its toxicity in PD [15].

2.3 LLPS of α -synuclein

Normally under physiological conditions in-vitro α -synuclein does not undergo LLPS, possibly because of an auto-inhibited conformational state [24]. But in certain conditions such as high local concentrations [10, 36], exposure to metal ions [10, 29, 32, 36, 37] or lipids [10, 14, 30, 36], pH changes [31, 36, 37] and oxidative stress [36] the protein can form liquid droplets. Under this disease associated conditions monomeric α -synuclein can transition into a phase-separated state and eventually form amyloid fibrils (**figure 4**) [36, 38]. Larger droplets are more prone to form amyloid fibrils, with the aggregation starting at the center of the droplets and then propagating to fill the entire droplet [39]. Alternatively, when already aggregated α -synuclein enters the condensates it initiates fibrillation by growing needle like extensions [40].

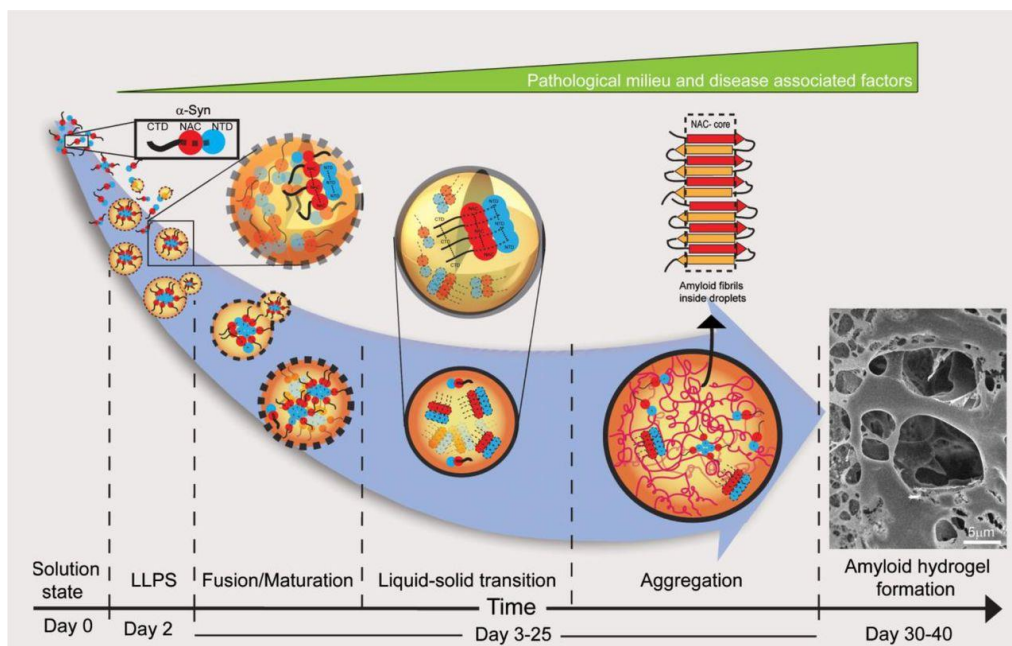


Figure 4. Schematic representation of α -synuclein phase separation and aggregation. α -synuclein transition from monomers into phase separated droplets. This transition then leads to the maturation of droplets from a liquid to a solid-like state through processes such as fusion and Ostwald ripening (which is roughly the growth of larger droplets at the expense of smaller ones) [27]. Within these droplets, α -synuclein condensates progressively become more rigid, eventually forming an amyloid hydrogel state composed of fibrillar aggregates and oligomers. The transition from liquid to solid-like phase is primarily driven by intermolecular interactions involving residues in the N-terminal and NAC domains of α -synuclein. Source: [35]

One paper theorized that the NAC region of α -synuclein gets shielded from interacting with other proteins by the N- and C-terminal domain. Through analysis of cross-link mass spectrometry (XL-MS), they were able to quantify the intra vs intermolecular interactions of the protein. It showed that before LLPS α -synuclein forms hairpin like monomers where the positively charged N-terminal and negatively charged C-terminal engage in long range electrostatic interactions, thus shielding the NAC domain. When the protein undergoes LLPS it then gets elongated, and the NAC domain is able to have more interprotein interaction (**figure 5A**) [41]. Another study looked into this in

the case of salt concentration (which was found to increase LLPS significantly over 500 mM) using Nuclear Magnetic Resonance (NMR) and Förster Resonance Energy Transfer (FRET) (**figure 5B**) [37]. Relevant to this hypothesis is a study on truncated C-terminal α -synucleins that found reducing the charge of the C-terminal increases the rate of phase separation [42].

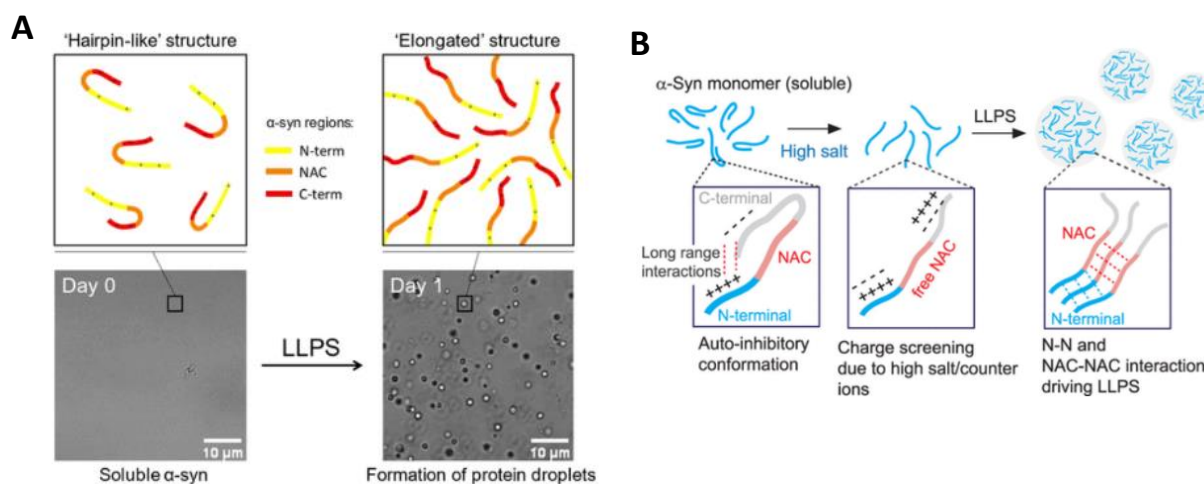


Figure 5. Schematic depiction of structure change in LLPS of α -synuclein. (A) Schematic showing the conformational change of α -synuclein that induces LLPS by freeing up the NAC domain together with microscopic images of the droplets. (B) Depiction of the interactions between the domains that cause LLPS to be promoted by salt concentration. Source A: [40], B: [36].

2.4 Methods in LLPS research

Our research will use optical microscopy to look at their size and shape and FTIR and circular dichroism to investigate secondary structure elements. There are many techniques and factors involved in LLPS research, in order to cover them properly it is helpful to have a general picture of the objectives in these studies. In order to research LLPS you first have to induce the phase separation. This is generally done for α -synuclein by using crowding agents such as PEG or dextran. After achieving droplet formation, one must analyze how the droplets behave, grow and potentially fuse with each other. This can be done with a wide variety of techniques, such as such as differential interference microscopy, electron microscopy, confocal microscopy and optical microscopy (**figure 6**) [32].

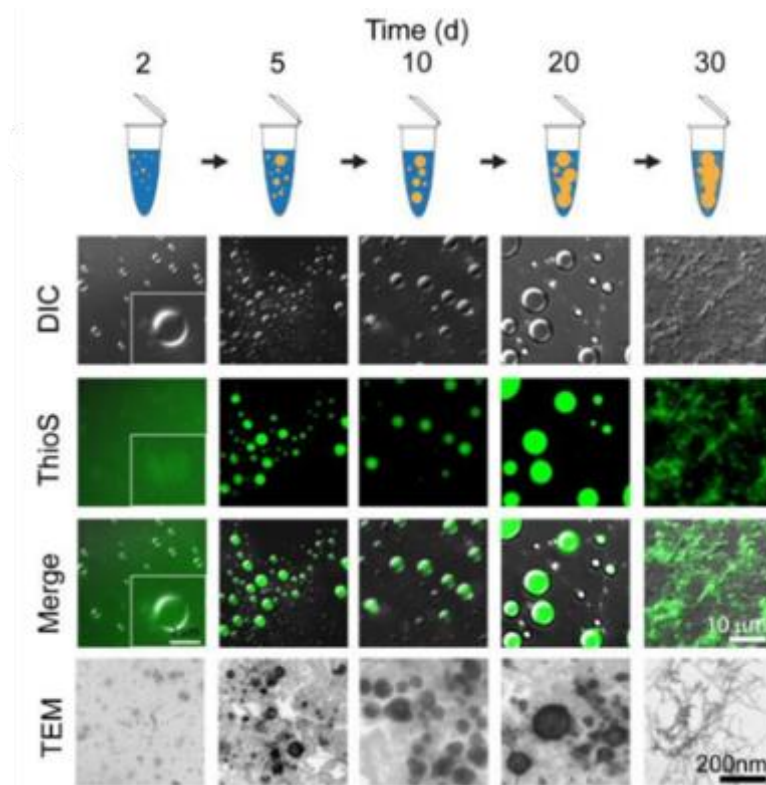


Figure 6. Example analyses of droplets over time. In this study researchers used Differential Interference Contrast Microscopy (DIC), ThioS staining, a combination of fluorescence images (Merge) and Transmission Electron Microscopy (TEM) to visualize and analyze droplet behavior over time. Source: [30]

2.4.1 General LLPS Induction

Many factors can influence LLPS in α -synuclein (**figure 7**) [38]. So, researchers often compare the effects of factors that are often associated with Parkinson's pathology. These include; exposure to ions such as Cu^{2+} , Fe^{3+} [10, 29, 32, 36, 37], NaCl [37], Mn^{2+} [43] or Ca^{2+} [30, 37], lipids [14, 30, 36], lowering the pH [31, 36, 37], mutations [14, 32-34, 36, 39, 44] and protein concentration of α -synuclein but also effects of other proteins such as tau [14, 34, 45-47], prion [48] and SERF [29] with some other tested conditions including purification methods [37], truncation of C-terminal domain [42], tRNA and polyU RNA [45], polypeptides [49], concentration of crowding agents like PEG [36] and biomolecules such as spermine [47]. Conditions and substances that slow down LLPS and/or aggregation (like acetylation [37], myricetin [35] and curcumin [33]) are of course also researched.

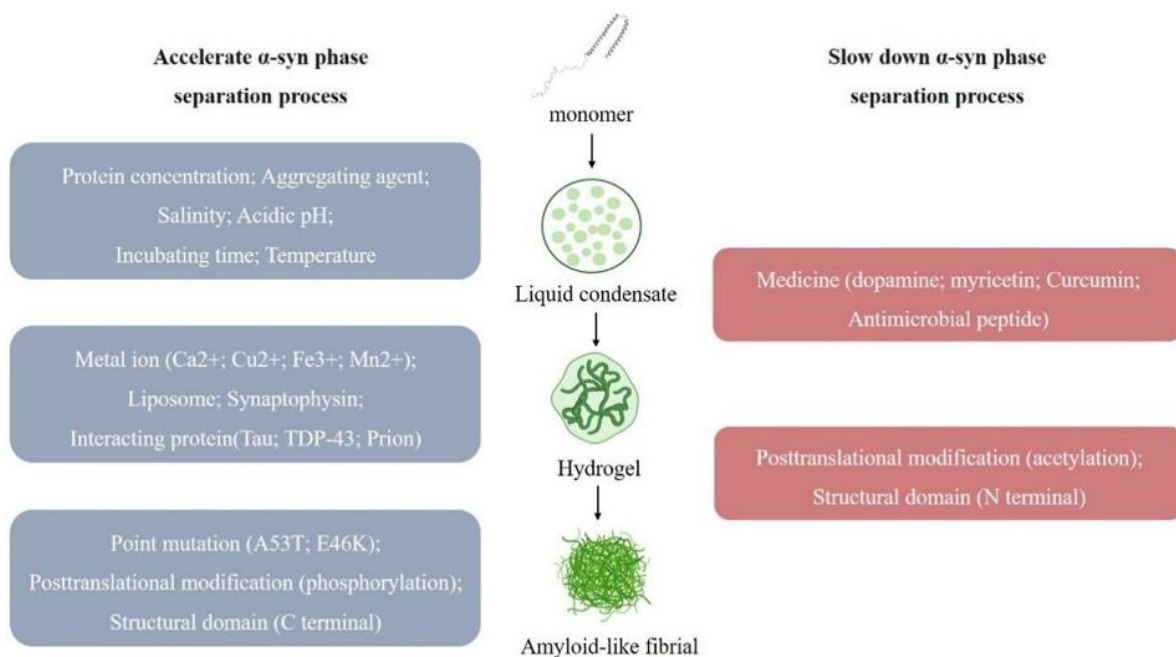


Figure 7. Several variables influence the process of α -synuclein LLPS. Several factors facilitate α -syn LLPS: (i) Experimental conditions, including protein concentration, presence of aggregating agents, salinity, acidic pH, incubation time, and temperature. (ii) External substances, such as metal ions (Ca^{2+} , Cu^{2+} , Fe^{3+} , Mn^{2+}), liposomes, synaptophysin, and interacting proteins (Tau, TDP-43, Prion). (iii) Intrinsic changes, such as point mutations (A53T, E46K), posttranslational modifications (phosphorylation), and structural domains (C-terminal). Conversely, various factors impede α -syn LLPS: (i) External substances, including medications (dopamine, myricetin, curcumin, antimicrobial peptides). (ii) Intrinsic changes in α -syn, such as posttranslational modifications (acetylation), mutations and structural domains (N-terminal). Source: [37]

PEG is a crowding agent that is used in most relevant experiments to induce LLPS of α -synuclein (table 1). Unfortunately, it can interfere with measurements, especially structural ones such as FTIR and CD; so, achieving and studying LLPS without PEG at physiological like conditions is important. Some studies use different crowding agents like dextran or positively charged polypeptides [50, 51] but these would have similar drawbacks. Exact conditions that can induce LLPS but remain physiologically realistic are needed.

Rey et al [36] showed promising results. At pH 7.4 (20 mM phosphate buffer) and after incubation at 37 degrees they managed to get LLPS droplet without PEG with an α -synuclein concentration of 500 μM . This concentration is much too high to be practical (the protein is very expensive) and it is far from physiological concentrations. They showed lowering the pH to 5.4 significantly lowers the threshold for phase separation as they achieved LLPS at 10 μM of α -synuclein without PEG. They also formed droplets using factors associated with fibrillation like Cu^{2+} , Fe^{3+} (50 μM) and liposomes (1 mM) with 200 μM α -synuclein in absence of PEG. They also they achieved LLPS with protein concentration as low as 5 μM with 100 μM Cu^{2+} and 1 mM liposomes.

Huang et al [30] showed that 2 mM Ca^{2+} also increases LLPS of α -synuclein although they did not test it without PEG. They also showed liposomes increased LLPS in combination with Ca^{2+} .

Table 1. Conditions to induce phase separation in α -synuclein research. Entries that are left blank were not specified in the research.

A-syn conc. (μ M)	PEG conc. (% w/v)	PEG MW (g/mol)	Buffer used	pH	Miscellaneous conditions and noteworthy information	Source
200	10	8000	20 mM sodium phosphate (NaPhos)	7.4		[10]
2	3	8000			6 μ M of EGFP reconstituted synapsin condensates induced LLPS or α -synuclein	[16]
30, 300	2, 5	8000	20 mM NaPhos			[29]
200	20	10000				[30]
250	10	8000		6.0 and 7.4		[31]
200	10	8000	20 mM NaPhos	7.4		[32]
200	20	10000				[33]
200	20	10000				[34]
200	20	10000			50 mM NaCl	[35]
10-500	10-20	8000	20 mM NaPhos	5.4 and 7.4	0.01% sodium azide	[36]
10-500	10 and 20	8000	20 mM NaPhos	5.5 and 7.4	0.01% sodium azide	[37]
200	10	8000	20 mM NaPhos	7.4	0.05% sodium azide	[39]
	0	NA			emGFP-5Fm protein induced α -synuclein LLPS	[40]
	10	8000	20 mM NaPhos	7.5	200 μ M peptides	[49]
200	15	10000				[42]
200	20	10000				[43]
500		8000		7.4	Could not achieve LLPS without using polyU and tRNA	[45]
	10	6000	35 mM tris-HCl	7.4	150 mM NaCl and 1 mM TCEP	[46]
10-250	0-20	6000	25 mM tris-HCl		0.01% sodium azide	[47]
30	0	NA			LLPS was induced by increasing salt concentration an unspecified amount	[48]
200	10	8000	20 mM NaPhos	7.4	0.01% sodium azide	[52]

0-100	Used other crowding agents: hexandiol (10%), Ficoll (10%) or Tween 80 (1%)	NA	100 mM Phosphate citrate, 20 mM phosphate or 20 mM PBS.	5, 5.5 6, 7 and 7.4	0, 50, 150, 300, 600 and 1000 mM NaCl was explored with 10 μ M sodium azide	[53]
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2.4.2 Fourier Transform Infrared (FTIR) and Circular Dichroism (CD) spectroscopy

Fourier-transform Infrared spectroscopy (FTIR) is often used in combination with circular dichroism (CD) [31, 36, 37], which can confirm the structural transition in proteins. CD is a spectroscopic method which depends on the fact that certain molecules interact differently with right and left circularly polarized light [54]. Using a cuvette containing the dissolved protein you can measure the change in orientation of the polarized light in millidegrees. This change in orientation is due to the sample absorbing left and right polarized light differently which has to do with its chirality. The spectrum can then be used to identify if the sample is mainly random coils (like in amorphous protein) or also contains α -helices and β -sheets with β -sheets being more frequent in aggregated protein.

Fourier Transform Infrared (FTIR) spectroscopy is a powerful analytical technique used to study the composition and structure of molecules, including proteins [55]. In FTIR spectroscopy, infrared radiation is passed through a sample, and the resulting absorbance or transmission of the radiation is measured as a function of wavelength [55]. These spectra can be used to determine differences in secondary structure by calculating the distribution of β -sheet, random-coil and turn/loop motifs [39, 56].

The basic principle behind FTIR spectroscopy lies in the interaction of infrared radiation with chemical bonds within the molecule. Different types of chemical bonds vibrate at characteristic frequencies in the infrared region of the electromagnetic spectrum. When infrared radiation of the appropriate frequency interacts with a molecule, it causes the bonds to absorb energy and undergo characteristic vibrational transitions. These transitions correspond to changes in the dipole moment of the molecule, resulting in absorption of specific wavelengths of infrared light [55].

In the context of protein structure analysis, FTIR spectroscopy can provide valuable information about the secondary structure, conformational changes, and interactions of proteins. The amide bonds in the protein backbone contain a C=O vibrational mode that is sensitive to the protein's secondary structure elements [55].

Quantitative analysis of FTIR spectra involves deconvoluting the complex spectra into individual components corresponding to different secondary structure elements. This can be achieved using spectral decomposition algorithms (Fourier Transform) and reference databases of known protein structures [55].

In the context of LLPS and α -synuclein research FTIR is not a common technique, but it has been used to characterize the distribution of beta-sheet, random-coil and turn/loop motifs [39, 56] (**figure 11**). Overall, FTIR spectroscopy offers a non-destructive and label-free method for probing the structural properties of proteins.

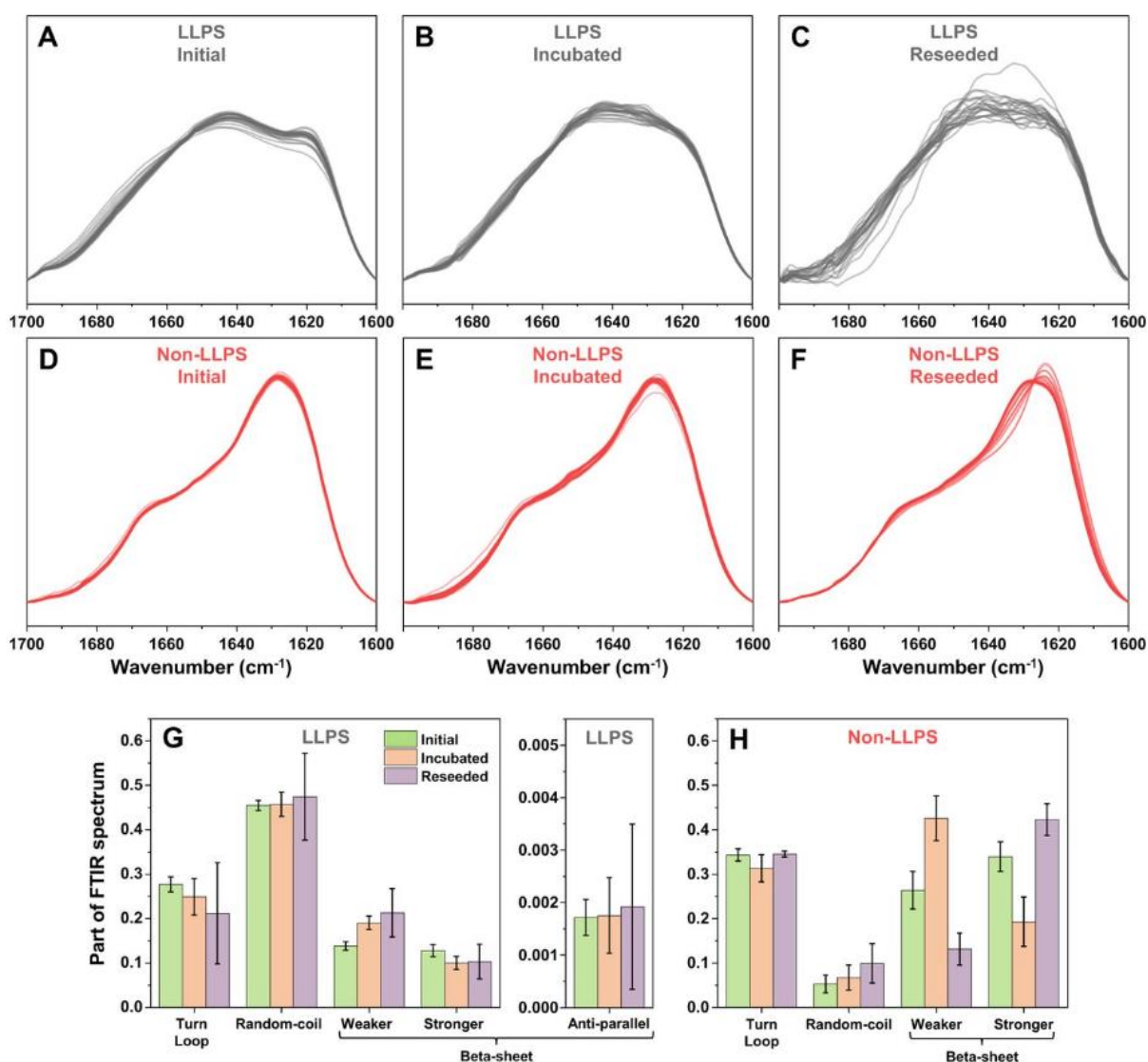


Figure 11. Example FTIR spectra and decomposition: LLPS and non-LLPS α -synuclein aggregate secondary structure distributions under different conditions. FTIR spectra of initial (A, D), cold-incubated (B, E), reseeded (C, F) LLPS and non-LLPS aggregates respectively. Turn/loop, random-coil and beta-sheet motif distribution of LLPS (G) and non-LLPS (H) samples, based on FTIR spectra deconvolution. Source: [59]

Building upon the techniques and conditions outlined above, our investigation into the liquid-liquid phase separation (LLPS) of α -synuclein involved a meticulous approach to the induction of phase-separated droplets. By employing optical microscopy, FTIR, and circular dichroism (CD) spectroscopy, we aimed to gain insights into the structural dynamics and behavior of α -synuclein under various conditions. The following results section details the outcomes of these experiments, highlighting the key observations and challenges encountered in our pursuit to understand the intricate dynamics of phase separation of α -synuclein.

3. Results

3.1 α -synuclein handling

α -synuclein requires careful handling since conditions such as shear forces and contamination can significantly impact its LLPS. Additionally, it is a very expensive material. Only small amounts of α -synuclein (~0.2 mg) are used, and due to its light weight, it can easily be displaced, even by a slight breath during weighing. The α -synuclein was carefully placed in the center of a small tube, ensuring that it did not touch the upper part where the lid meets the tube. If too much α -synuclein was weighed, it could not be returned to the stock vial to prevent potential contamination of the entire stock. After accurate weighing, the α -synuclein was dissolved in the buffer by gently pipetting the solution up and down for a few seconds, taking care to avoid inducing aggregation through shear forces. On the day 0 measurements the sample was left to stand for ~30 minutes before starting measurements. When taking sample solution for measurements the appropriate volume is first carefully pipetted up and down 3 times to mix and ensure a representative measurement. Three different α -synuclein stocks were used: an older untagged α -synuclein from Cambridge, UK, a newer α -synuclein from Merck, and an α -synuclein from Anaspec that was labeled with Hilyte FluorTM 488.

3.2 Microscopy based analysis of α -synuclein

To investigate phase separation, different samples of α -synuclein were analyzed using confocal, brightfield, and fluorescence microscopy techniques. First we tried to reproduce the results of [36], which were able to induce LLPS of α -synuclein at pH 5.5 and 5.4 in the presence of 5-20% PEG with α -synuclein concentration down to 5 μ M. Since droplets are not homogeneous, with variations in environment and particle flow between the edge and center, it was important to capture images from multiple positions within each droplet. For instance, evaporation at the edges can increase the local concentration of particulates, which is why images were taken at the edge, middle, and halfway points of the droplet. Additionally, particles tend to settle at the bottom where the droplet meets the well plate, a factor that is particularly relevant for confocal microscopy when viewing a z-slice of the droplet.

Given these considerations, the first measurement for each sample was taken in a higher plane above the surface of the droplet where it contacts the well plate, using a confocal microscope. The sample would be scanned from 50 to 800 microns above the surface to find objects of interest). Because condensates can be extremely small, often in the sub-micron range [35], high-resolution imaging was necessary. Thus, 0.08 microns per pixel with 1024 pixels at 60x magnification was used for confocal microscopy, and 0.07 microns per pixel with 2048 pixels at 100x magnification was used for widefield microscopy.

α -synuclein supplied by Cambridge was initially tested at pH 7.4 and 5.4 in PBS buffer but produced little to no condensates with a few contaminants (**figure 12**).

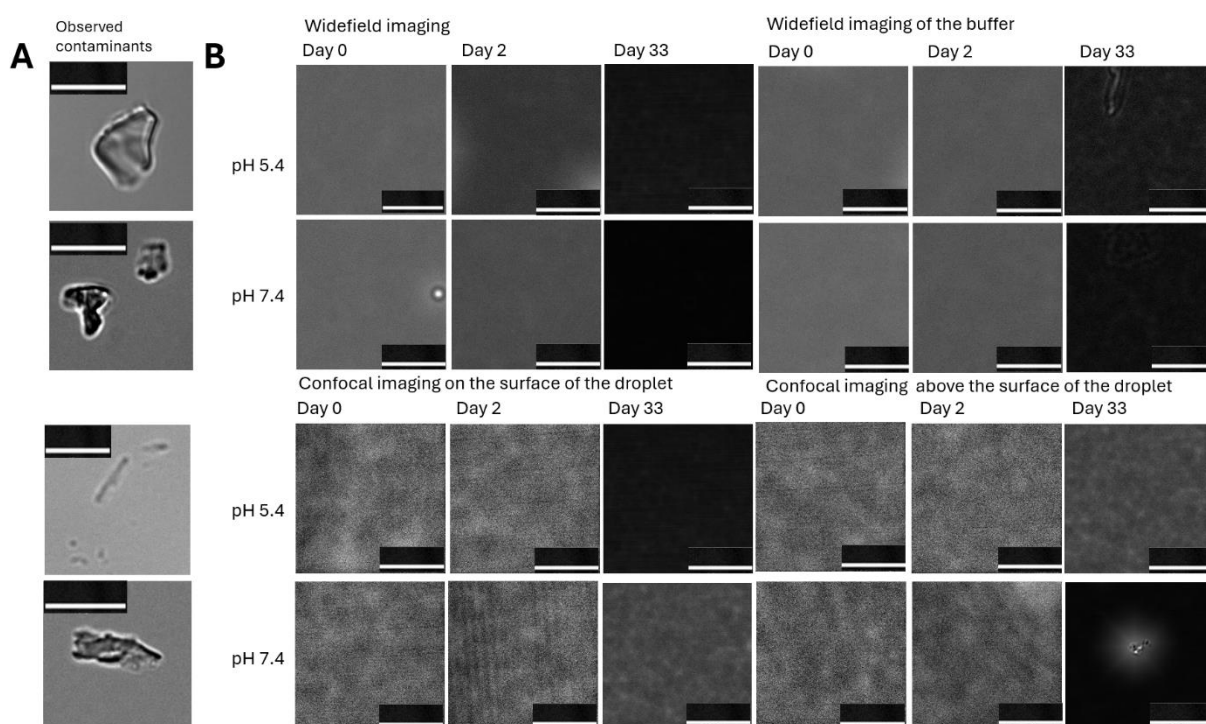


Figure 12. Optical analysis of α -synuclein in PBS buffer at pH 5.4 and 7.4 at 20 μ M protein concentration (all scale bars are 10 microns, in this and every other figure in the results). (A) Contaminants, though rare, appear in all samples, including the buffer. (B) Confocal optical imaging on and above the sample substrate surface and optical brightfield imaging on day 0 and after 2 and 33 days of incubation at 37 $^{\circ}$ C.

According to what was found in literature phase separation should occur in these conditions, especially at pH 5.4 [36, 37]. The lack of phase separation could be due to the protein being old. Therefore, we continued with a different, newer α -synuclein supplied by Merck. This α -synuclein was used under the same conditions and imaged similarly. After 6 days of incubation, fluorescently labeled α -synuclein was added in a 1:10 ratio to the unlabeled α -synuclein to better visualize potential condensates. The concentration was also measured using a nanodrop spectrophotometer (**figure 13**).

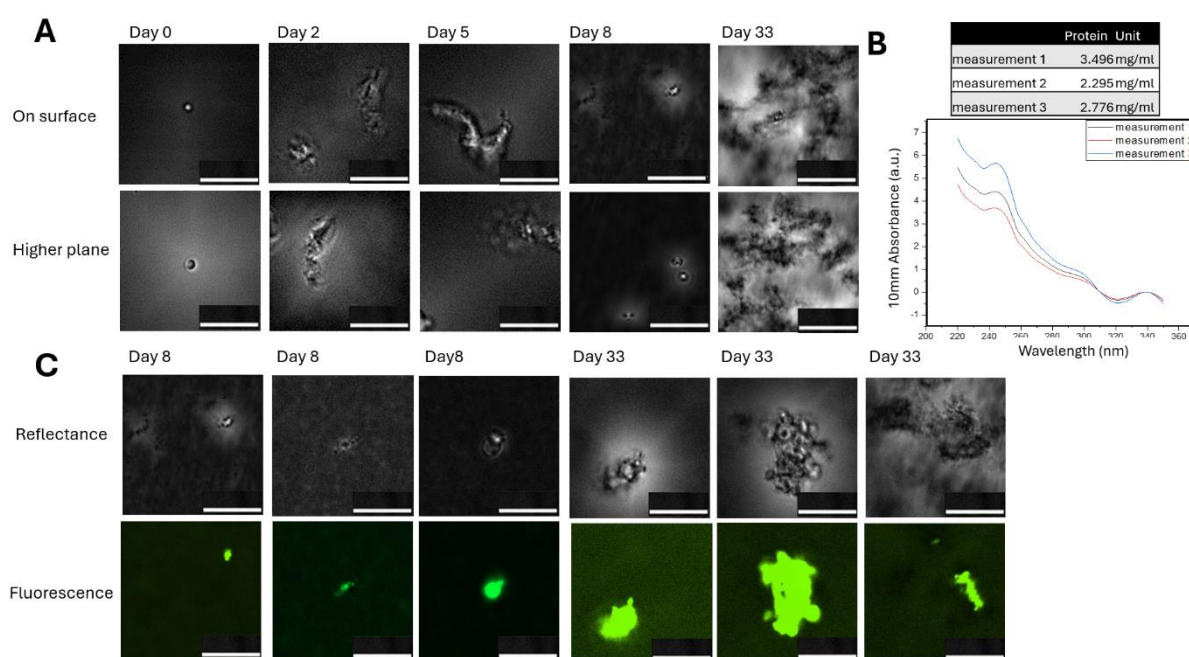


Figure 13. Analysis of the second α -synuclein assay in PBS buffer at pH 5.4, at 70 μ M protein concentration. (all scale bars are 10 microns). (A) Optical confocal imaging at different time points after incubating at 37 $^{\circ}$ C. (B) Fluorescent confocal imaging of protein structures. (C) Nanodrop analysis of the protein sample.

Although this sample exhibited more observable structures, the behavior still did not align with expectations. Structures with different and disordered shapes were noticed with higher frequency compared to the first experiment after the fluorescent dye was added, though their fluorescent decreased in the last measurement on day 33. Some of these structures align with the expected size and shape of condensates but their number was not in agreement with results from literature [36]. Over time, irregular non-fluorescent aggregates formed, even though α -synuclein is expected to aggregate much more slowly [36]. These could be caused by impurities in the sample, alternatively contaminants in the PEG could also be a cause as the PEG we used was very old. The nanodrop shows inconsistent and incorrect concentrations and its spectrum is also irregular. Normally there is a peak at 280 nm, it is likely that the fluorescent label is interfering even at this wavelength and drowning at the protein absorbance. It could also have something to do with possible protein impurities.

The experiment was repeated with the fluorescently labeled α -synuclein added from the start instead of later on. This time, the protein appeared to produce aggregates much earlier, though again, they were not fluorescent (**figure 14**).

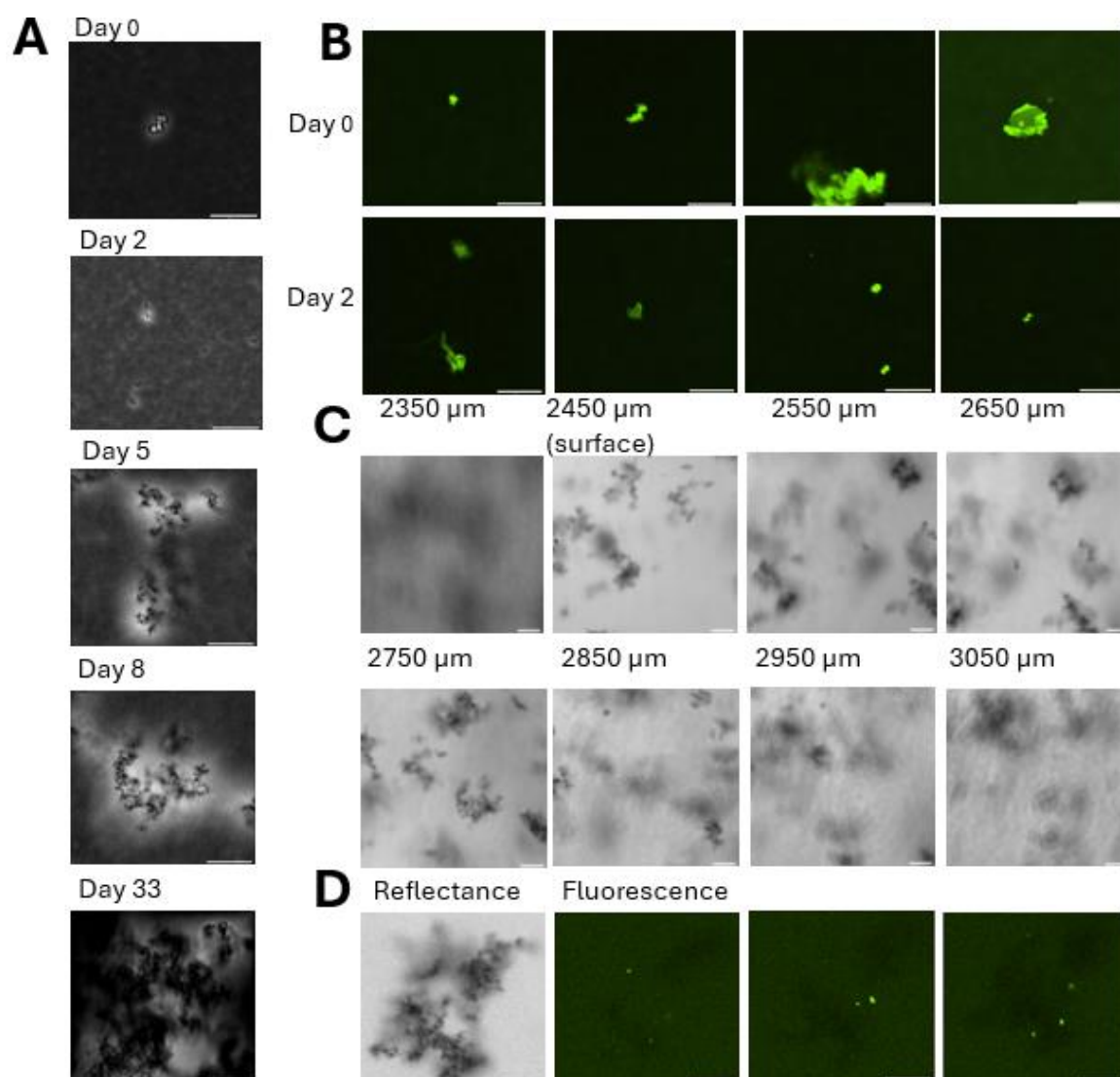


Figure 14. Analysis of the third LLPS assay of α -synuclein in PBS buffer at pH 5.4, again at 70 μ M α -synuclein concentration (all scale bars are 10 microns). (A) Development of large structures over time. (B) Fluorescent structures on day 0 (top row) and after 2 days of incubation (bottom row). (C) Large scale imaging of structures along the z-axis of the sample. (D) Mostly non-fluorescent structure in confocal reflective and corresponding fluorescent mode, showing different fluorescent components in slightly different z planes.

The same phenomenon of cloudy non-fluorescent aggregate-like structures appears in this sample. But rather more quickly than in the previous experiment. Unusual fluorescent structures of α -synuclein were observed initially which could be explained by fluorescent α -synuclein attaching to contaminants perhaps. These structures diminished as aggregates became more prominent. These “aggregates” occurred less higher above the surface of the droplet like you would expect from more solid structures. Another peculiar observation is the presence of small fluorescent parts on edges the aggregates. This could be droplets of α -synuclein that are trapped inside the aggregates.

3.3 Investigation of absence of LLPS: Spectroscopic techniques

Suspensions arose that something was wrong with the protein from Merck. Notably, the vial contained much more than the stated 1 mg of α -synuclein. Furthermore, the behavior of the α -synuclein under the microscope did not align with literature, the samples were expected to produce fluorescent condensates from day 0 [36]. To determine whether the structures were aggregates or something else, and to assess the purity of the protein, Nanodrop, FTIR, and circular dichroism analyses were performed (**figure 15**).

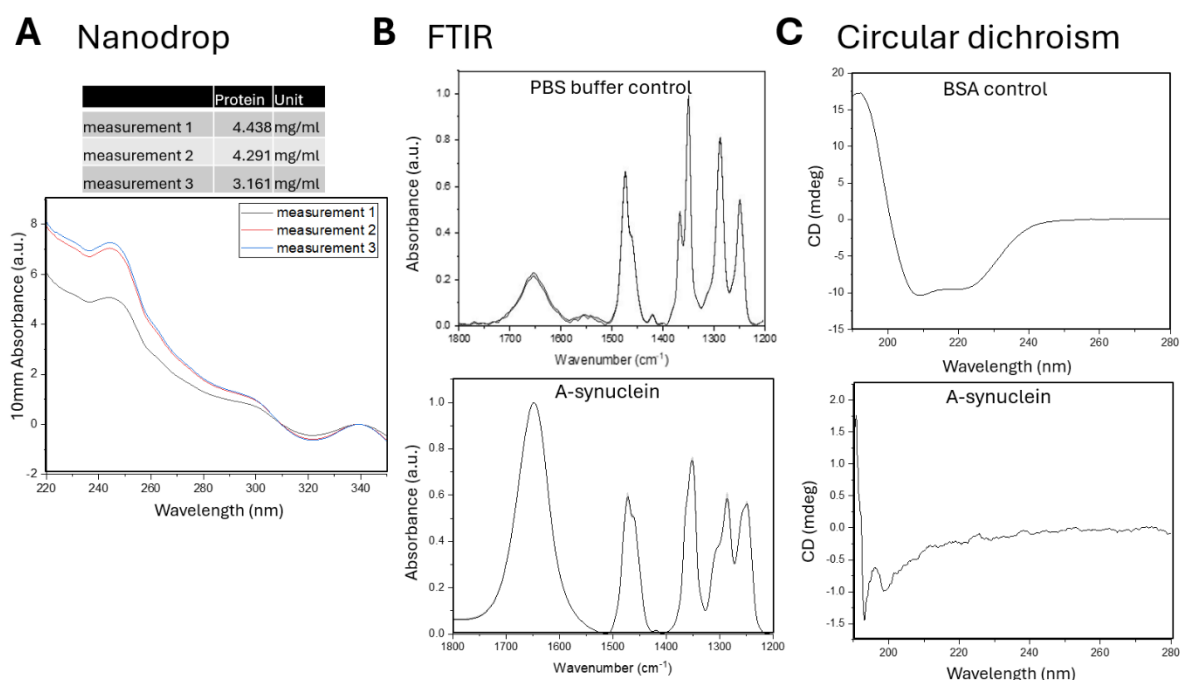


Figure 15. (A) Nanodrop protein concentration with associated spectrum showing inconsistent and incorrect concentrations. (B) FTIR spectrum of PBS buffer control (top) and α -synuclein sample after 8 days of incubation (bottom). Buffer spectrum indicates PEG interference. (C) Circular dichroism spectra of positive control BSA at 0.5 mg/mL (top) and α -synuclein at 1 mg/mL (bottom) both with a 0.1mm path length, showing no significant signal above noise for α -synuclein but significant signal for the positive control.

For the nanodrop spectrum you would expect a sharp descent at 240 nm becoming relatively flat with a peak around 280 nm. However, this was not observed and it showed inconsistent values. This could again be due to the fluorescent label emitting enough to drown the 280 nm protein peak. The FTIR spectrum appeared unusual, potentially due to PEG interference, as the buffer spectrum exhibited similar peaks, likely attributable to the high PEG concentration. The characteristic peaks of PEG reported in the literature do not fully align with our findings [57], but the peak around 1475 cm^{-1} could still be attributed to the O–H vibrations, while the lower peaks may correspond to the CH_2 vibrations of PEG. The FTIR spectra was not deconvoluted to investigate the secondary structure elements as the spectrum does not align with a normal protein spectrum which should not contain so many strong peaks below 1400

cm^{-1} . The circular dichroism spectrum shows very little signal. This could be because the PBS buffer decreases the signal due to a scattering effect.

To further test, two new samples without PEG, containing unlabeled and labeled α -synuclein at 70 and 7 μM respectively, were analyzed using a Nanodrop device, FTIR, and CD (figure 16).

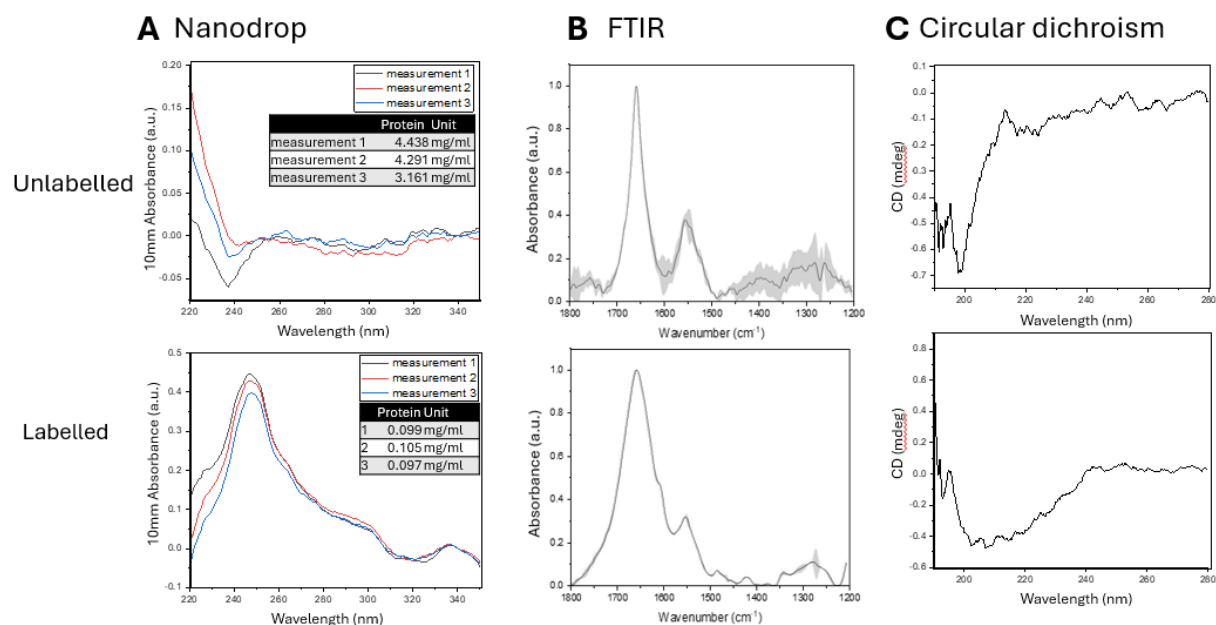


Figure 16. Analysis of labeled (7 μM) and unlabeled (70 μM) α -synuclein on day 0 in the absence of PEG. (A) Nanodrop spectra with associated measured protein concentration, showing 0.1 mg/mL as expected for the labeled α -synuclein but incorrect values for the unlabeled α -synuclein. (B) Averaged FTIR spectra showing two amide peaks at ~ 1650 and ~ 1550 associated with protein. (C) Circular dichroism spectra that are not significantly above noise.

Intrinsically disordered proteins (IDPs) like α -synuclein lack a stable secondary structure in their native state, leading to distinct spectroscopic signatures. In Fourier Transform Infrared (FTIR) spectroscopy, IDPs typically exhibit sharp amide I bands around 1655 cm^{-1} . Similarly, in Circular Dichroism (CD) spectroscopy, intrinsically disordered proteins (IDPs) typically exhibit a characteristic spectrum that reflects their lack of stable secondary structure. IDPs generally show a strong negative peak around 200 nm, which is indicative of a random coil structure. This negative peak arises because the random coil conformation does not impart significant optical activity to the circularly polarized light, leading to minimal differential absorption. In contrast, well-defined secondary structures such as α -helices and β -sheets produce distinct CD signals: α -helices show negative peaks near 208 nm and 222 nm with a positive peak around 190 nm, while β -sheets exhibit a negative peak near 218 nm and a positive peak around 195 nm. The minimal signal in these regions for IDPs confirms the absence of significant α -helical or β -sheet content, reinforcing their predominantly disordered nature.

The Nanodrop analysis of the unlabeled α -synuclein showed incorrect concentrations, but there was no interference from a fluorescent label this time, as none was present.

The spectrum looks like a spectrum of a sample containing too little protein to measure. The labeled α -synuclein sample exhibited similar interference in the Nanodrop spectrum as before, although the concentration measurements were accurate. The FTIR of the unlabeled α -synuclein looks very good and so it is clear there is protein in there. The labeled FTIR has a broader amide peak which is atypical for a IDP, this could be caused by the fluorescent dye absorbing in that range. What was strange is that the FTIR spectrum of the labeled α -synuclein had a signal 10 times higher (~ 0.01 at the amide 1 peak) than the spectrum of the unlabeled one (~ 0.001). This is especially peculiar as the labeled α -synuclein has the lower concentration in this case (7 μ M vs 70 μ M). To explain this contradiction we hypothesized that the Merck α -synuclein was not very pure and contained a lot of non-protein. The circular dichroism spectra are both at a very low signal so it is not a reliable metric here but the unlabeled α -synuclein spectrum does show a random coil pattern indicating non-aggregated monomeric protein. The labeled α -synuclein meanwhile looks more like an α -helix spectrum.

It seems the purity of the unlabeled α -synuclein powder might not be 95% as stated on its data sheet. To further test the unlabeled α -synuclein, a new sample with a very high concentration of 2.38 mM (or 34 mg/mL) was made (**figure 17**).

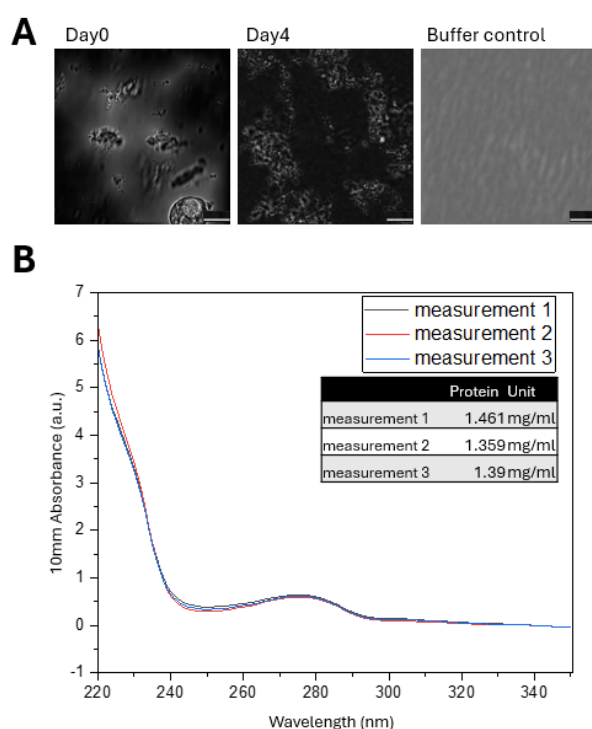


Figure 17. Investigation of Merck supplied unlabeled α -synuclein in PBS buffer containing 20% (w/v) PEG at pH 5.4 at high protein concentration (34 mg/mL or 2.4 mM) (all scale bars are 10 microns). (A) Images of α -synuclein before and after 4 days of incubation at 37 $^{\circ}$ C as well as the buffer control showing the protein is highly contaminated. (B) Nanodrop spectrum showing a good protein curve with a peak around 280 nm but indicating concentrations much

The concentration used in this experiment, while intentionally high, exceeds the typical solubility limit of α -synuclein under standard conditions. Such elevated concentrations are generally impractical, as they push the protein beyond its solubility threshold, increasing the likelihood of precipitation or aggregation. Despite these challenges, the imaging results revealed the presence of contaminants, which were distinct from any undissolved or aggregated protein. This observation offers a plausible explanation for the unusual structures observed in previous experiments, where the presence of

extraneous particles may have confounded the interpretation of phase separation and aggregation behavior.

Interestingly, while the sample confirmed the presence of α -synuclein—evidenced by a characteristic Nanodrop peak—the measured concentrations were significantly lower than expected. This discrepancy suggests that a substantial portion of the protein may have either precipitated out or failed to dissolve properly, further complicating the analysis. Evidence also points to the protein not being pure and containing a significant amount of non-protein. This would explain why we never observed condensates as the protein concentration was just too little. Filtering the protein could help to reduce the presence of observed contaminants

Given these findings, and in an effort to minimize variables that could obscure the true behavior of the protein, subsequent experiments were conducted exclusively with the labeled α -synuclein. This approach aimed to ensure better visualization and consistency, reducing the impact of any potential contamination or solubility issues (figure 18).

3.4 Reassessment of LLPS or α -synuclein with microscopy

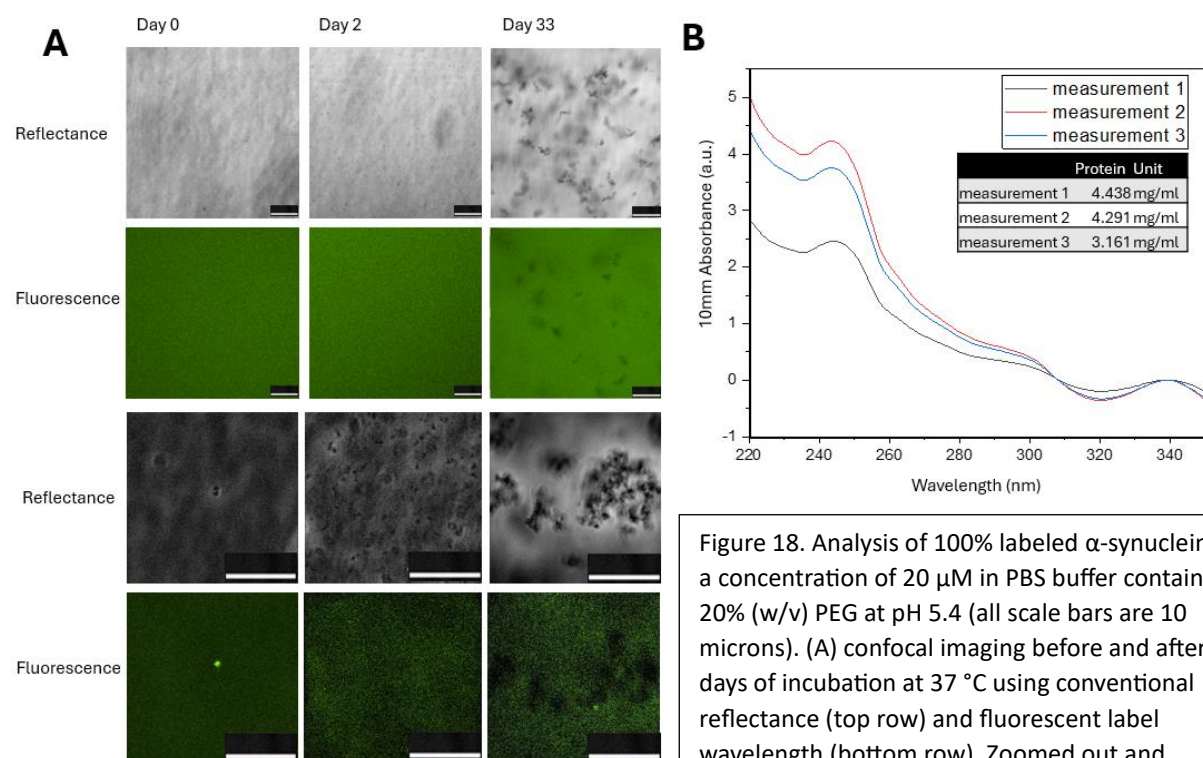


Figure 18. Analysis of 100% labeled α -synuclein at a concentration of 20 μ M in PBS buffer containing 20% (w/v) PEG at pH 5.4 (all scale bars are 10 microns). (A) confocal imaging before and after 2 days of incubation at 37 °C using conventional reflectance (top row) and fluorescent label wavelength (bottom row). Zoomed out and zoomed in images are compared. (B) Nanodrop spectrum on day 0 showing incorrect concentrations indicating inference from fluorescent label.

Initially, the fluorescently labeled α -synuclein appeared to produce condensates that increased in number over time, but these were not visible with fluorescence. This could

be because they are too small or not fluorescent enough to show up above the noise of the monomeric α -synuclein. It is also possible that the condensates were moving too fast for the microscope to capture the fluorescence accurately. However, by day 5, the same non-fluorescent, cloudy structures appeared that were seen in earlier experiments. This indicates that they were not caused by contaminants in the unlabeled α -synuclein but rather that they have something to do with the labeled α -synuclein. The nanodrop spectrum, like in the other samples that contain fluorescently labeled α -synuclein, shows incorrect concentrations with an unusual spectrum, likely due to inference from the fluorescent label.

PEG may be contributing to the formation of the cloudy structures, potentially interacting with the fluorescent label in some way. To further investigate, the next experiment involved optical imaging of α -synuclein in the absence of PEG and with a reduced concentration of labeled α -synuclein, adjusted from a 1:10 to a 1:200 ratio (**figure 19**).

3.5 Mitigating contaminants and refining buffer conditions to induce LLPS of α -synuclein

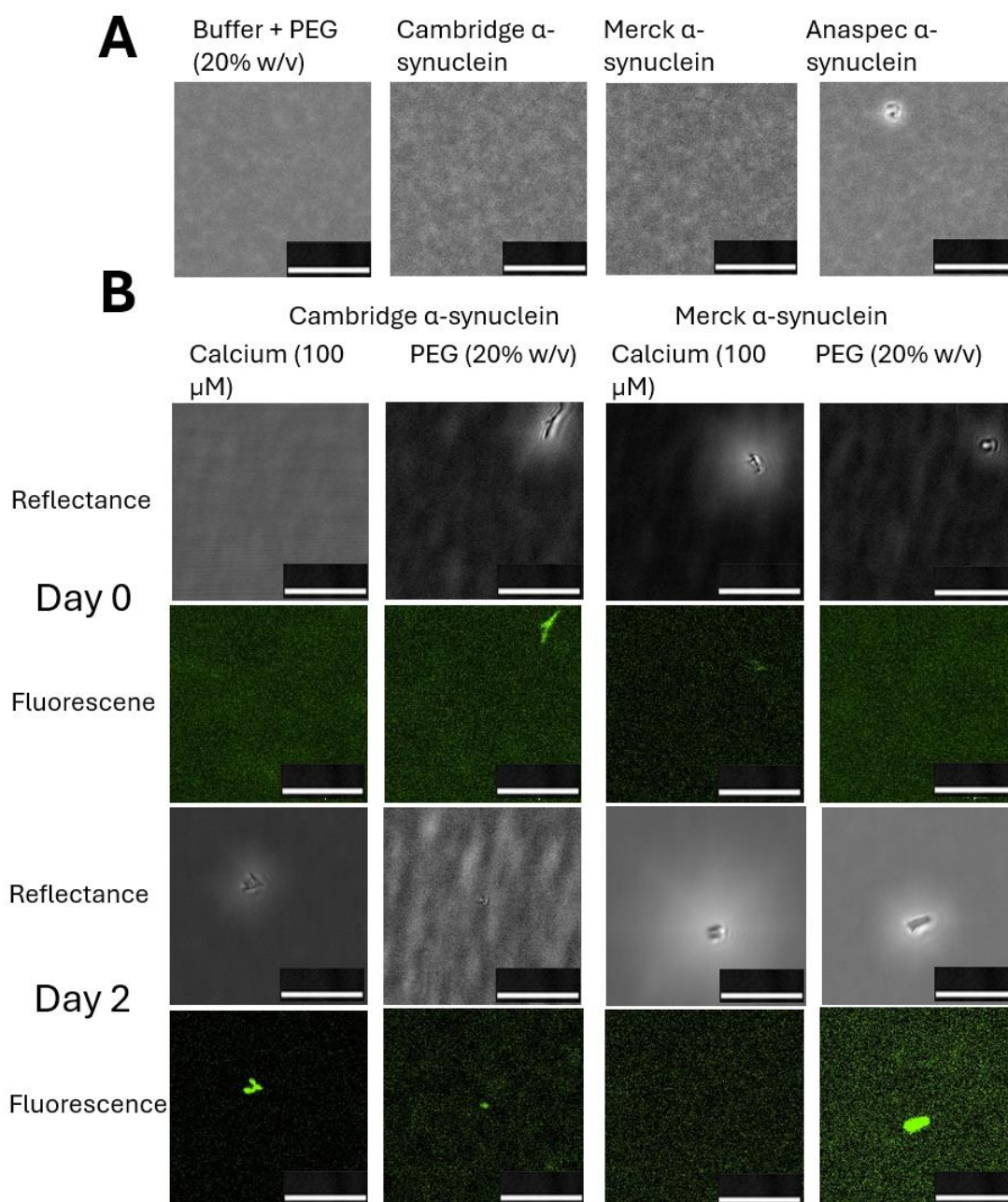


Figure 19. Analysis of α -synuclein from different suppliers after filtering with a 100 kDa filter at pH 5.4 in PBS buffer (all scale bars are 10 microns). (A) Widefield check for contaminants of PEG and the 3 α -synuclein stocks at 70 μ M without PEG. (B) Confocal images comparing Cambridge and Merck supplied α -synuclein at 140 μ M protein concentration in the presence of calcium of PEG before and after 2 days of incubation at 37 $^{\circ}$ C. Labeled Anaspec supplied α -synuclein was present in a 1:200

To address the potential contamination and further investigate the effects of buffer conditions on LLPS, additional experiments were conducted using a 100 kDa filter to remove contaminants from all samples. We also examined the role of PEG and calcium by systematically varying buffer conditions, and analyzed α -synuclein behavior in the

absence of PEG to ensure a more controlled environment. 100 μ M calcium was used instead of PEG for some of the samples as it has been shown to be inductive of phase separation in α -synuclein [30].

After filtering with the 100 kDa filter, the buffer exhibited significantly fewer contaminants, as did the α -synuclein samples. However, the Anaspec-supplied α -synuclein, which could not be filtered due to its initial 70 μ M concentration in solution, still showed some contaminants, though these were infrequent. When Merck and Cambridge α -synuclein samples were analyzed at higher concentrations post-filtering, no phase separation was observed in the case of using PEG as well as calcium. Although some contaminants were present, they were not significant, indicating that the previous observations of phase separation were likely due to contamination rather than intrinsic protein behavior.

These rigorous tests and analyses provide insights into the complexities of α -synuclein behavior, highlighting the challenges of studying LLPS and aggregation in vitro. Understanding the precise conditions under which α -synuclein undergoes phase separation or forms aggregates is essential for unraveling its role in neurodegenerative diseases like Parkinson's. These findings underscore the importance of eliminating contaminants and optimizing experimental conditions to draw more definitive conclusions about the protein's behavior.

4 Conclusion & Discussion

In this study, we aimed to investigate the induction of liquid-liquid phase separation (LLPS) in α -synuclein using various protein samples and bulk techniques. Our experiments highlighted several key findings.

Firstly, the older, untagged α -synuclein from Cambridge exhibited negligible phase separation, likely due to the long storage time. This emphasizes the importance of protein freshness in LLPS experiments. However, the Merck α -synuclein, although new, also demonstrated peculiar behavior. The labelled α -synuclein from Anaspec provided better visualization and more consistent results, underlining the efficacy of fluorescent labeling in tracking LLPS. It also seemed to cause non-fluorescent aggregate-like structures over time, which was atypical for α -synuclein. Discrepancies in Nanodrop and CD measurements indicated significant contamination, especially with the Merck sample. Microscopy analysis of a high concentration sample confirmed heavy contamination. This suggests that stringent purity checks are crucial for reliable LLPS studies. FTIR spectra showed significant contribution from PEG, which makes it challenging to perform direct structural analysis due to the bulk nature of this technique. This calls for careful consideration of buffer components in future experiments.

Furthermore, our rigorous tests using a 100 kDa filter and varying conditions, including the use of calcium and a higher protein concentration, revealed that previous observations of phase separation were likely influenced by contamination rather than intrinsic protein behavior. This finding emphasizes the importance of eliminating contaminants and optimizing experimental conditions to draw more accurate conclusions about α -synuclein's LLPS and aggregation behavior.

Our study faced several challenges that limited our ability to dive deeper into the complex behavior of α -synuclein. Originally it was intended to use fluorescence after photobleaching (FRAP) and Thioflavin-T-fluorescence in a plate reader to investigate the transition from liquid droplets to solid fibrils. Additionally AFM could have been used to look at the droplets on the nanoscale and identify the minimum nucleus size for LLPS to occur. To do these techniques however, it was first crucial to reliably induce phase separation which was challenging in this thesis. In future research these techniques should be applied as well as compare LLPS in more conditions, varying salt and protein concentration, pH and excluding PEG.

By addressing these areas, future research can make significant strides in understanding α -synuclein LLPS and its role in disease, ultimately contributing to the development of targeted therapeutic interventions.

5 Experimental protocol

5.1 Buffer preparation

A buffer solution containing 20 mM Na₂HPO₄, 150 mM NaCl, 20% (w/v) polyethylene glycol (PEG) 8000, and 0.005% (w/v) sodium azide was prepared. The reagents were measured using aluminum weighing dishes and a 10 mL pipet for the water. The components were added to a 100 mL glass beaker as follows:

Component	Volume (mL)	Weight (mg)	Molecular Weight (g/mol)	Concentration	Pot no.
Na ₂ HPO ₄	0.239	141.96	141.96	20 mM	208955
NaCl	0.902	438.3	58.44	150 mM	612644
PEG-8000	11.25	10000	8000	20% (w/v)	642120
Sodium azide	0.009	5	-	0.005% (w/v)	511461
milliQ water	36.6	-	-	-	-

Table 2. Buffer components

A small amount of sodium azide was rinsed from the aluminum plate with 1 mL of MilliQ water, included in the 36.6 mL listed. The beaker was stirred with a magnetic stirrer for approximately 20 minutes to dissolve all components, covered with a clean aluminum plate to prevent contamination.

The pH was adjusted from approximately 8 to 5.4 (or 7.4 for the first experiment) using HCl and NaOH solutions while continuously stirring. The final volume was brought up to 50 mL with MilliQ water. The buffer solution was filtered through a 0.45 µm membrane filter using a 5 mL syringe and transferred to a 50 mL Falcon tube. A 0.6 mL aliquot was further filtered through a 0.2 µm PFTE membrane filter using a 1 mL syringe and placed in a 1.5 mL Eppendorf tube. Contaminants were checked using a confocal microscope by placing a 10 µL droplet on an 18-well Ibidi plate. If contamination was clear from viewing the buffer optically it would be filtered again. But sometimes there would still be a significant amount of contaminants. In this case the buffer was prepared again from scratch, being extra careful and clean.

For the last experiment calcium was added instead of PEG. This was done by first dissolving calcium chloride at 10 mM in 20 mM Tris-HCl buffer at pH 7.4 as described in literature [37]. This is done to prevent the calcium chloride from reacting with the sodium phosphate.

5.2 α-Synuclein Preparation

Upon confirming minimal contaminants in the buffer, recombinant α-synuclein was retrieved from a freezer (-20 °C in the case of Merck supplied α-synuclein and -80 °C for the Cambridge α-synuclein). Using cleaned spatulas (washed with tap water, MilliQ water, and ethanol, and wiped with soft tissues), around 0.2 mg of α-synuclein was placed into a 0.2 mL PCR tube on a 0.01 mg accurate scale. The protein was dissolved in the appropriate amount of filtered buffer solution to achieve the desired concentration. In case labeled α-synuclein was used it was also taken from -80 °C bio storage and added in the appropriate amount. It would then be refrozen using snap freezing with liquid nitrogen and returning to bio storage. 25 µL aliquots of the labeled

α -synuclein were made to prevent unnecessary freeze and thaw cycles which can damage the protein. After everything was added the sample was mixed gently by pipetting up and down to avoid bubbles, and left at room temperature for 30 minutes before continuing with imaging or other measurements.

For filtering the α -synuclein solutions, which was done in the last experiment, a 100 kDa amplicon filter was used. To facilitate the solution going through the filter it was centrifuged at 6000 RPM for around 1 hour. Although centrifugation can cause aggregation through shear forces, α -synuclein aggregates more slowly and has been shown to be resistant to centrifugation [58].

5.3 Optical microscopy

For imaging, 10-20 μ L of the α -synuclein solution was carefully deposited onto an 18-well Ibidi plate using 100 μ L Greiner Bio-One tips. Unlike some other studies that sandwich the droplet between two glass plates [35, 36, 38], this method may influence phase separation by altering the surface area of the glass interface. After deposition, the remaining solution was sealed with parafilm, and in the case of fluorescent labels, aluminum foil was also used to protect the sample. The samples were then incubated at 37°C.

Confocal imaging was performed using a Nikon C2 confocal laser scanning microscope with Nis-elements software. Widefield microscopy was conducted with a Zeiss fluorescence microscope using ZEN blue-pro software, though a Nikon Eclipse Ti2 inverted microscope was also used once, again with Nis-elements software. For the confocal microscope, a 60x objective was used, providing a resolution of 0.08 microns per pixel across 1024 pixels, with the pinhole set to its minimum of 1.2 AU. For optical widefield microscopy, magnifications of 60x to 100x were employed, yielding a resolution of 0.07 microns per pixel across 2048 pixels, with the intensity display set to auto.

Imaging involved capturing pictures at various positions within the droplet, including near the edges, the middle, and intermediate areas. For confocal imaging, additional images were taken at higher planes above the droplet's surface. When fluorescence was involved, both non-fluorescent and fluorescent images were captured for each area.

The images were processed using ImageJ software, where they underwent background subtraction with the sliding paraboloid option, and smoothing was disabled. Brightness and contrast were adjusted slightly if necessary, images were cropped, and a scale bar was added.

5.4 Fourier Transform Infrared (FTIR) spectroscopy Analysis

Fourier Transform Infrared (FTIR) spectroscopy was performed using a Tensor 2 Bruker device with DTGLS detector and a diamond ATR with temperature control and OPUS software. Background measurements were taken before each sample measurement. A 0.5 μ L droplet of α -synuclein solution was deposited and allowed to dry for approximately 8 minutes. Subsequently, 5 μ L of MilliQ water was used to rinse the protein by pipetting it sideways over the dried droplet to remove buffer components. The surrounding area was gently dried with a soft paper towel, and after another 8 minutes of drying, measurements were conducted. Each measurement consisted of 200 scans, repeated three times to obtain three averaged spectra from three different droplets.

The spectra obtained from the FTIR machine were processed in OriginPro 2022 software. The spectra were cut from a 4000 to 400 nm size to only 1800 to 1200 which

contained the peaks that were most relevant. Then the peaks were smoothed, first using adjacent averaging with a window size of 3 and secondly using Savitzky-Golay with a window size of 7. The graph is then baselined, using points chosen for each graph individually so no value goes below 0. The Y-axis is normalized to go from 0 to 1 and as a last step the graph is averaged from the 3 technical repeats with the standard deviation added as well.

5.5 Circular Dichroism (CD) Spectroscopy

Circular dichroism was conducted using a J-815 spectropolarimeter equipped with a temperature controller and Spectra Manager software. The instrument was flushed with nitrogen gas for approximately 15 minutes before measurements. A buffer blank was first measured using carefully cleaned cuvettes (washed with ethanol and MilliQ water). Measurements were conducted using 0.1 mm and 1 mm path length cuvettes, requiring 35 and 300 μL volumes, respectively. The α -synuclein sample would be measured in the same cuvette as the buffer after thorough cleaning. The measurements were taken over a spectra range of 280 – 190 nm with a data pitch of 0.1 nm. The bandwidth was set to 2 nm and the speed of the measurement to 100 nm/min. The samples were stabilized at a temperature setpoint of 25 °C. Each measurement comprised 10 accumulations, averaged to produce one spectrum. After protein measurements a final cleaning was done using 1% Helmanex which aims to make sure no protein is left behind on the glass.

For analysis the spectrum of the buffer blank was subtracted from the sample spectrum in Excel. The data was then copied to origin and graphed

5.6 Nanodrop

In order to check protein concentrations a nanodrops spectrophotometer device was used. 1.5 μL droplets were deposited on the device after thorough cleaning with ethanol. First the buffer is used as a blank, then it's cleaned and after which the sample is deposited and measured 3 times. To have an accurate measurement the device software (Nanodrop 2000) needed to know the extinction coefficient at 280 nm ($5960 \text{ M}^{-1} \text{ cm}^{-1}$) and molecular weight (14.46 kDa) of the α -synuclein.

After spectrum acquisition the data was processed in Excel. The data was then copied to origin and plotted.

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